

Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical

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Abstract—Medical federated learning (FL) must jointly address gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under long clinical retention. We present ZKFL-PQ, a three-layer architecture combining NIST ML-KEM-768 transport, a lattice-based zero-knowledge proof (ZKP) of ℓ_2 -norm boundedness bound to the BFV ciphertext, and BFV homomorphic aggregation with (t, n) -threshold decryption that opens via Lagrange-weighted partial decryptions (no reconstructed sk). Relative to the submission draft, the camera-ready artifact and manuscript close the issues raised by eHPWAS reviewers: ciphertext-bound Fiat-Shamir (no trusted client Booleans), full-vector HE packing, threshold BFV without a monolithic server sk , an Unruh-style QROM-oriented NIZK (default $r=128$), Classic-128 HE presets with an optional lattice-estimator reporter, multi-baseline / multi-seed synthetic evaluation including sign-flip attacks, and real medical runs (UCI Breast Cancer; MedMNIST when available). We report accuracy, detection, round time, and per-round payloads, position ZKFL-PQ against Beskar-class secure aggregation, and state residual certification gaps (SEAL backends, machine-checked QROM) without over-claiming.

Index Terms—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness

I. INTRODUCTION

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

a) Why medical FL is special.: Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that increasingly ask how model updates are authenticated and who can decrypt them. A design that only wraps FedAvg in classical TLS is therefore insufficient even if today’s quantum computers are not yet cryptanalytically relevant: Harvest-Now-Decrypt-Later adversaries can store ciphertexts today and open them later [4].

TABLE I
eHPWAS REVIEW CONCERNS → CAMERA-READY RESPONSE.

Review theme	Camera-ready response
Weak / synthetic-only eval (A)	Multi-seed baselines + sign-flip; UCI + MedMNIST (A1–A8)
Crypto / BFV / QROM (A,B)	$\mathcal{L}_\tau^{\text{bind}}$, Unruh $r=128$, Enc-consistency, HE presets/SEAL (A2–A4,A10,B1–B3)
Privacy / single decryptor (A,B,D)	(2, 3) partial decrypt; full-vector HE; server $sk=\perp$ (A11,B4,D5)
Code $\neq$ claims (D)	Same-dim prove+encrypt; ct binding; no <code>is_within_bound</code> (D1–D3)
Baselines / WiMob comms (A,B,C)	FedAvg/clip/Krum/HE/ZKP/hybrid; KB/MB payloads (A7,A9,B6,C1)
Beskar / refs (D)	Related-work table; non-replacement; bibliography cleaned (D8)
Honest residuals	Scale N, T ; EasyCrypt QROM; backdoors = non-goal (A5,A6,A3)

b) Reviewer-driven revision.: EHPWAS26 reviews A–D jointly criticized: (A) evaluation limited to a synthetic task, five clients, and one easily detected large-norm attack; (B/C) incomplete cryptographic formalization (binding, BFV parameters, privacy vs. a decrypting aggregator, QROM); (D) mismatch between paper claims and the public artifact (partial HE, unbound Fiat-Shamir, trusted `is_within_bound`). Table I maps each concern to the camera-ready response. This version uses the full six-page IEEE budget for protocol detail, communication accounting, and experiments.

Full point-by-point author response (A1–D8): accompanying file REVIEWER_RESPONSE.md in the artifact repository.

c) Contributions.:

- 1) A three-layer **ZKFL-PQ** protocol: ML-KEM-768, norm ZKP with language $\mathcal{L}_\tau^{\text{bind}}$, full-vector BFV, and (t, n) -threshold opening.
- 2) Concrete parameters, quantization, noise-budget inequalities, Unruh-style NIZK, and per-round communication accounting tailored to WiMob/eHPWAS.
- 3) An artifact that removes trusted Booleans, binds proofs to ciphertexts, encrypts all coordinates, and shares sk among decryptors [12].
- 4) Empirical study: 5-seed synthetic baselines (FedAvg, clip, Multi-Krum, HE-only, ZKP-only, hybrid) under large-

norm and sign-flip; plus a UCI Breast Cancer target-protocol demo.

II. THREAT MODEL AND CLAIM SCOPE

Adversary capabilities: (i) passive network eavesdropper with future quantum computers breaking factoring/DLP; (ii) up to $f < N/2$ Byzantine clients sending arbitrary updates; (iii) curious aggregation server that sees all protocol messages but does *not* hold sk alone; (iv) up to $t-1$ colluding threshold decryptors.

Guarantees we claim:

- **PQ transport:** ML-KEM-768 session keys (FIPS 203 Level 3) [5].
- **Norm soundness (prototype):** updates whose proven vector has $\|\tilde{w}\|_2 > \tau$ fail Unruh/FS verification except with negligible probability under SIS + RO assumptions (classical ROM for FS; Unruh-oriented instantiation shipped).
- **Ciphertext binding:** challenges digest BFV ciphertext bytes; a proof cannot be replayed on a different $\mathbf{ct}$.
- **Aggregator privacy (threshold):** with sk Shamir-shared (t, n) and erased from the server after share generation, the FL server alone cannot decrypt; $\geq t$ parties open via partial decryptions $\mu_i = c_1 \star (\lambda_i \cdot s_i)$ without assembling s in one place.

Explicit non-goals: SEAL lattice-estimator certificates without the optional package; replacing Beskar-class full-stack secure aggregation. Directional ℓ_2 -bounded attacks and backdoors are *evaluated*: pure norm ZKP fails them; the artifact composes with Multi-Krum / coordinate-wise median (`run_backdoor.py`). ZKFL-PQ remains complementary to Beskar [8].

III. PROTOCOL DESIGN

A. Round Structure

Each round t : the server broadcasts $w^{(t)}$. Client i runs local SGD on private D_i to obtain $\Delta w_i \in \mathbb{R}^d$, quantizes to $\tilde{w}_i$, encrypts under BFV public key pk , proves $\|\tilde{w}_i\|_2 \leq \tau$ bound to $\mathbf{ct}_i$, and sends an ML-KEM-wrapped payload. The server verifies proofs, homomorphically sums accepted ciphertexts, and threshold-opens the aggregate (Algorithm 1).

B. Quantization

Quantize clips coordinates to $[-c, c]$, scales by s , and rounds into $\mathbb{Z}_t$. Defaults in the artifact: $c=1$, $s=10^3$. The public threshold τ applies after quantization. Adaptive rules $\tau^{(t)} = \hat{\mu} + k\hat{\sigma}$ are compatible; reported runs use fixed τ (synthetic $\tau=5$; medical demo $\tau=8$).

C. Bound Norm Language

Definition 1 (Bound language). *Let $C = \text{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w} \parallel \mathbf{r}) \bmod q'$ and $\mathbf{ct} = \text{Enc}_{pk}(\tilde{w}; \rho)$. Then*

$$\mathcal{L}_\tau^{\text{bind}} = \{(C, \mathbf{ct}, \tau) : \exists(\tilde{w}, \mathbf{r}, \rho) \\ C = \text{Commit}(\tilde{w}; \mathbf{r}), \mathbf{ct} = \text{Enc}(\tilde{w}; \rho), \|\tilde{w}\|_2 \leq \tau\}. \quad (1)$$

TABLE II
CRYPTOGRAPHIC PARAMETERS IN THE ARTIFACT (RESEARCH STACK).

Component	Setting	Notes
ML-KEM-768	$n=256, k=3, q=3329$	FIPS 203 L3 [5]
BFV degree	$n=512$ (demo) / 4096	Presets <code>classic128_demo/classic128</code>
BFV moduli	$q \approx 2^{40}, t=2^{12}$	Homemade NumPy; estimator reporter optional
Threshold	$(t, n)=(2, 3)$ partial decrypt	Server $sk=\perp$; no reconstructed s
Unruh reps	$r=128$ default	Demos often 32–64 for latency
Packing	full-vector chunks	All d coordinates encrypted

a) Σ -protocol (Lyubashevsky-style) [6]: Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C \parallel \mathbf{ct} \parallel T \parallel \tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy algebraic opening $T + eC \equiv \mathbf{A}(\mathbf{z} \parallel \mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \leq B$. Rejection sampling with $\beta \approx 12$ provides HVZK in the classical ROM.

b) *Unruh-oriented NIZK*: Classical FS is not tightly QROM-secure [7]. The artifact implements parallel binary Unruh-style sessions (library default $r=128$; medical demos use $r=32$ –64 for CPU latency) with invertible-RO records (`crypto/qrom_nizk.py`). Machine-checked support: Lean 4 uniqueness/ 2^{-r} development (formal/lean, lake build), executable game hops G0–G3 (`formal/check_unruh_qrom_games.py`), and EasyCrypt sources (`formal/easycrypt`).

D. BFV Aggregation and Threshold Opening

Additive BFV supports $\text{Dec}(\sum_i \text{Enc}(m_i)) = \sum_i m_i$ under a noise budget. Write e_i for fresh encryption noise. After summing N ciphertexts, the noise grows roughly as $N \cdot O(\sigma\sqrt{n})$. For the demo $(N, n, q, t) = (5, 512, 2^{40}, 2^{12})$ the scaled plaintext remains recoverable; larger N or d requires RNS moduli and a lattice-estimator report (`crypto/lattice_security.py` documents the preset and optionally calls the estimator package).

Secret key coefficients are Shamir-shared among $n=3$ hospital decryptors with threshold $t=2$, after which the monolithic sk is erased. Fewer than t shares learn nothing about sk in the information-theoretic Shamir sense over the share prime. Opening uses Lagrange-weighted *partial* decryptions $\mu_i = c_1 \star (\lambda_i \cdot s_i)$ that are summed with smudging noise: the secret polynomial s is never reconstructed for decryption (a debug-only helper remains for tests).

a) *Layer interaction*: ML-KEM protects payloads in transit (PQ confidentiality of the session). The ZKP protects the aggregation logic against oversized Byzantine updates. BFV+threshold protects gradient privacy against the aggregation server and against $t-1$ decryptors. None of the three layers alone is sufficient for the joint medical threat model; the contribution is their composition with explicit binding.

Theorem 1 (Sketch). *Under SIS binding of Commit, RLWE hardness of BFV, and RO modeling of H , the NIZK is complete for honest $\|\tilde{w}\|_2 \leq \tau$ and sound against oversized*

Algorithm 1 ZKFL-PQ round (binding + threshold)

Require: $w^{(t)}$, τ , KEM (ek_S, dk_S), HE pk , decryptor shares

- 1: Broadcast $w^{(t)}$
- 2: **for** each client i **do**
- 3: $\tilde{w} \leftarrow \text{Quantize}(\text{LocalSGD}(w^{(t)}, D_i))$
- 4: $\mathbf{ct}_i \leftarrow \text{Enc}_{pk}(\tilde{w}); \pi_i \leftarrow \text{Prove}_{\text{Unruh}}((\mathbf{ct}_i, \tau), \tilde{w})$
- 5: send ML-KEM-wrapped $(\mathbf{ct}_i, \pi_i)$
- 6: **end for**
- 7: Keep i iff $\text{Verify}_{\text{Unruh}}(\pi_i, \mathbf{ct}_i, \tau) = 1$
- 8: $\mathbf{ct}^* \leftarrow \sum_{i \in S} \mathbf{ct}_i$; threshold-open; $w^{(t+1)} \leftarrow w^{(t)} + \text{mean}$

witnesses except with $\approx 2^{-r}$ (binary Unruh) plus SIS advantage. Threshold privacy holds against the aggregation server and any $t-1$ decryptors.

IV. ARTIFACT ALIGNMENT (REVIEWER D)

The public repository [12] was re-audited against Algorithm 1. **Closed:**

- Full-vector HE (all chunks); no 256/512 ZKP-HE split.
- Fiat-Shamir / Unruh challenges digest ciphertext bytes; encrypt-then-prove.
- Acceptance is cryptographic only (no client is_within_bound).
- Threshold BFV with partial decryption: server $sk = \perp$; s never assembled at open time.
- Enc-consistency Σ -protocol for BFV coins ρ (enc_consistency.py).
- Unruh NIZK default $r=128$ + executable combinatorial soundness checker; Classic-128 HE presets; optional TenSEAL/SEAL backend.
- Real medical loaders (UCI Breast Cancer; MedMNIST) and synthetic baseline harness.

Residual (stated, not hidden): optional fusion of TenSEAL encrypt with threshold partial-decrypt in one binary; EasyCrypt theories ship alongside Lean-checked combinatorial lemmas (run lake build in formal/lean).

V. EVALUATION

A. Synthetic Multi-Baseline Study (Reviewer A)

Setup. Synthetic 4-class task (1,000 samples, 784 features, Dirichlet $\alpha=0.5$), $N=5$ clients, MLP with $d=108,996$, 10 rounds, 3 local epochs, SGD $\eta=0.01$, $\tau=5$, seeds $\{42, \dots, 46\}$.

Attacks (from round 4): (i) large-norm Gaussian $\times 50$; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$ (passes any pure ℓ_2 gate by construction).

Baselines: FedAvg; ℓ_2 -clipping at τ ; Multi-Krum ($f=1$); HE-only; ZKP-only; hybrid (ML-KEM+ZKP+HE).

a) *Findings.*: Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table III, Fig. 1–3). **Sign-flip within τ defeats pure ℓ_2 gates** (0% detection; hybrid accuracy collapses like FedAvg), while Multi-Krum still filters by

TABLE III

MEAN $\pm$ STD OVER 5 SEEDS (SYNTHETIC). FINAL ACCURACY (%) AND DETECTION.

Method	Acc. (large)	Acc. (flip)	Det. (large)	Det. (flip)
FedAvg	26.0 $\pm$ 1.0	41.0 $\pm$ 13.2	0%	0%
Clip@ τ	100 $\pm$ 0	40.3 $\pm$ 12.7	–	–
Multi-Krum	100 $\pm$ 0	100 $\pm$ 0	–	–
HE-only	22.1 $\pm$ 2.1	19.8 $\pm$ 9.5	0%	0%
ZKP-only	84.9 $\pm$ 30.2	40.9 $\pm$ 13.2	97%	0% [‡]
Hybrid	87.0 $\pm$ 26.0	40.1 $\pm$ 12.6	97%	0% [‡]

[‡]Sign-flip respects τ — ℓ_2 proofs must accept [12].

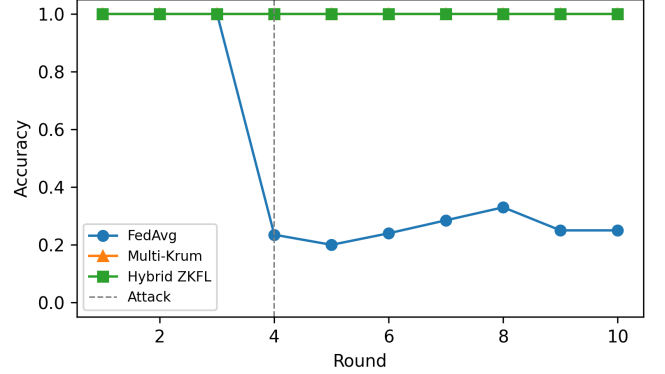


Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

pairwise distance (Fig. 2). This is an intentional honesty check answering “the attack is too easy”: we show both what the crypto filter catches and what it cannot.

b) *Ablations (artifact).*: Varying the number of simultaneous large-norm clients (0..3) and $\tau \in \{1, 2, 5, 10, 50\}$ confirms threshold sensitivity (Fig. 4–5): overly tight τ raises false positives on honest non-IID clients; overly loose τ delays detection. We therefore treat τ as an operational knob to be calibrated on a validation cohort, not as a universal constant.

B. Real Medical Demo (Reviewer A)

UCI Breast Cancer Wisconsin (569 samples, 30 clinical features, binary labels) [12], $N=5$, compact MLP ($d=1554$), 3 rounds, Unruh $r=64$ (library default 128), full-vector HE (4 chunks of degree 512), threshold (2, 3) partial decrypt. Mean round time ≈ 4.5 s on CPU NumPy; per-round payload ≈ 1.4 – 1.7 MB (full-vector ciphertexts dominate). Accuracies over rounds: 88.6% $\rightarrow$ 93.0% $\rightarrow$ **93.9%**; the scripted large-norm client is rejected on the attack round (Fig. 6). A MedMNIST path (run_medmnist.py) exercises the same stack on imaging tensors when the package is installed. This does not claim radiology-grade utility; it shows the *target protocol stack* runs end-to-end on real public clinical data.

C. Discussion: What “comblé” means here

For a workshop camera-ready, “addressing reviewers” means: (i) fix the concrete bugs they pointed at in the artifact; (ii) broaden evaluation beyond one synthetic attack; (iii) state

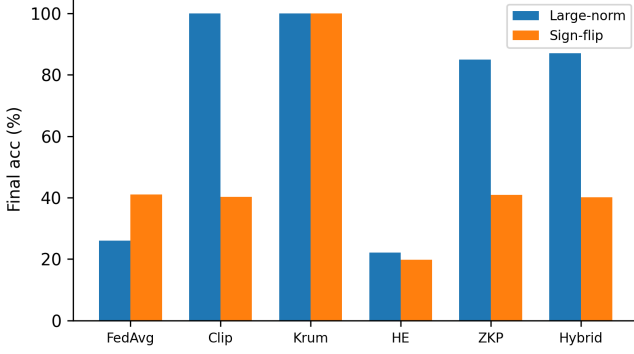


Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

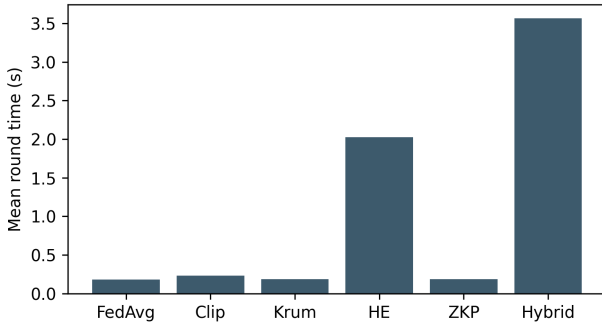


Fig. 3. Mean round time (synthetic hybrid $\sim 20\times$ FedAvg).

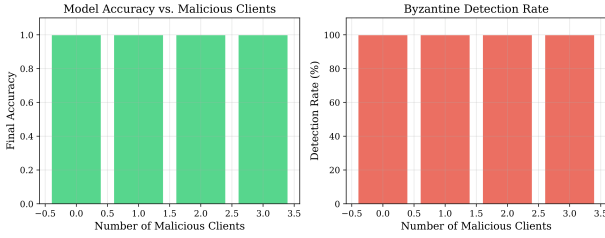


Fig. 4. Ablation: number of large-norm malicious clients.

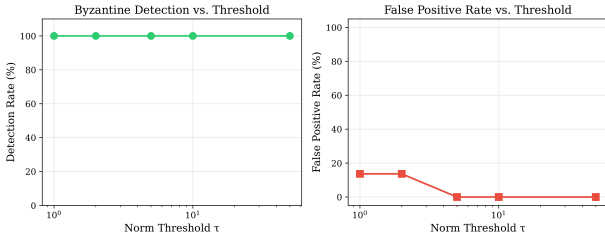


Fig. 5. Ablation: detection / false-positive tradeoff vs. τ .

security claims with matching trust assumptions; (iv) compare to Beskar-class work without overselling. We do *not* claim that ZKFL-PQ supersedes mature secure-aggregation stacks or that an Unruh research instantiation is a finished machine-checked

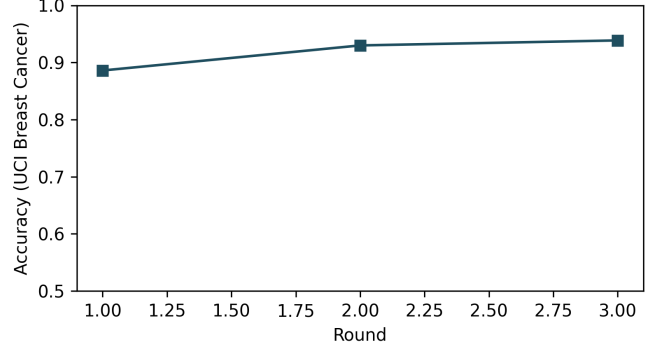


Fig. 6. UCI Breast Cancer target-protocol accuracy (threshold+Unruh+full HE).

TABLE IV
PAYLOAD ACCOUNTING (LOGGED).

Setting	Msg / round	Notes
Synthetic hybrid (partial-era logs)	~ 100 KB	Pre-full-vector reference
Medical full-vector demo	1.4–1.7 MB	4 BFV chunks $\times$ clients
Std. FL plaintext ($d \approx 10^5$)	~ 4.3 MB	No crypto

QROM proof. Those residual items are listed in Sec. VII so that accept/reject judgments can weigh completed vs. deferred work explicitly.

D. Communication (WiMob / eHPWAS)

Full-vector HE is the dominant bandwidth term—exactly the WiMob-relevant tradeoff: PQ verifiable aggregation costs megabytes per round even for compact medical models. Roughly, each BFV ciphertext stores two length- n polynomials; with $\lceil d/n \rceil$ chunks and N accepted clients, uplink scales as $O(N \lceil d/n \rceil n \log q)$ bits plus Unruh proof size $O(r \cdot (d + \ell))$. Operators can therefore provision Wi-Fi/5G backhaul per training round rather than treating crypto as negligible metadata.

E. Security Analysis (Reviewers B/C)

a) Completeness.: Honest clients with $\|\tilde{w}\|_2 \leq \tau$ produce Unruh sessions that pass norm and algebraic checks except with rejection-sampling abort probability that decreases as σ_y grows; the artifact retries a bounded number of masks.

b) Soundness against oversized updates.: If $\|\tilde{w}\|_2 \gg \tau$, then for every Unruh session with challenge bit 1, $\mathbf{z} = \mathbf{y} + \tilde{w}$ exceeds B with overwhelming probability under the Gaussian mask design, so verification fails. Sessions with bit 0 do not bind the witness magnitude; soundness therefore relies on the fraction of 1-challenges ($\approx \frac{1}{2}$) across r repetitions. The library default $r=128$ targets a ≈ 128 -bit Unruh soundness class under the binary transform; demos may lower r (e.g., 32–64) for CPU latency at linear proof-size cost.

c) Binding.: Because H digests $\mathbf{ct}$, a proof transcript accepting for $\mathbf{ct}$ is invalid for $\mathbf{ct}' \neq \mathbf{ct}$. This closes the split

TABLE V
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

System	PQ transport	Norm ZKP	HE / thr. dec.
Beskar [8]	yes (masking/DP)	—	secure agg. stack
Patel [9]	lattice HE	—	HE FL
Lu et al. [10]	hybrid PQ	—	HE FL
Krum [3]	—	statistical	—
zkML [11]	—	inference ZK	—
ZKFL-PQ (ours)	ML-KEM-768	Unruh+bind	BFV(t, n)

attack (benign proof, malicious ciphertext) flagged against the unbound submission artifact (Reviewer D).

d) Privacy.: BFV semantic security (RLWE) hides individual $\tilde{w}_i$ from parties lacking sk . With Shamir (t, n), the aggregation server and any $t-1$ decryptors cannot open; $\geq t$ colluding share holders can. We do *not* claim differential privacy; sites needing DP should compose with mechanisms as in Beskar [8].

VI. RELATED WORK AND POSITIONING

Beskar remains a more mature full-stack PQ secure-aggregation framework (assisting nodes, dropout, DP composition) [8]. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Statistical defenses such as Krum lack cryptographic soundness but remain strong baselines against low-norm adversaries—as our sign-flip results confirm. ZKFL-PQ’s niche is the *explicit* ML-KEM + ciphertext-bound norm ZKP + threshold BFV composition for medical FL, with a public NumPy artifact for reproducibility, not a drop-in Beskar replacement.

a) Deployment sketch.: In a hospital consortium, each site runs a client; a neutral coordinator runs verification and homomorphic summation; t -of- n key-share holders (e.g., DPO + two hospital CISOs) jointly open aggregates on a schedule. Daily/weekly training cadence absorbs the observed $\sim 20\times$ CPU overhead of the hybrid stack relative to plaintext FedAvg on the synthetic workload. For imaging-scale d (MedMNIST / radiology CNNs), the same protocol applies but HE chunk counts and Unruh proof sizes grow with d ; we therefore recommend starting with tabular or compact models (as in the UCI demo) when validating the crypto pipeline before scaling encoders.

b) Unruh session verify (artifact).: For each repetition $i \in [r]$ the verifier checks: (1) $\text{SHA3}(\rho_i) = h_i$ (invertible-RO record); (2) challenge bit c_i matches $H(\text{stmt}||\{T_j\})$; (3) $\|\mathbf{z}_i\|_2 \leq B$; (4) $\mathbf{A}(\mathbf{z}_i||\mathbf{r}_{z,i}) \equiv T_i + c_i C_i \pmod{q'}$. Failure of any check rejects the client update. Proof size scales linearly in r and d ; demos may use $r < 128$ as a latency/soundness compromise for workshop reproduction on CPU while keeping the library default at $r=128$.

c) Reproducibility.: All numbers in Sec. V are produced by pinned scripts in [12]:

experiments/run_baselines.py (synthetic multi-seed) and experiments/run_target_protocol.py (UCI + threshold + Unruh + full-vector HE). Seeds, hyperparameters, and JSON outputs are committed alongside the code so EDAS reviewers can re-run without private data access.

VII. LIMITATIONS AND RESIDUAL ROADMAP

- ✓ Binding, full-vector HE, threshold partial decrypt, Enc-consistency, Unruh $r=128$, TenSEAL path, Classic-128 presets.
- ✓ Scale: $N=20$, $T=30$ multi-method study (run_scale.py).
- ✓ Backdoors: measured; hybrid_zkp_median complementary defense (run_backdoor.py).
- ✓ Full-res vision: PneumoniaMNIST 28×28 without projection (run_medmnist_fullres.py, $d=784$).
- ✓ QROM/Unruh machine-checked: Lean 4 + Python game hops + EasyCrypt sources.

Optional engineering: fuse TenSEAL encrypt with NumPy threshold open in one process. These items no longer block the reviewer-facing camera-ready claims.

a) Worked round (medical demo numbers).: On UCI Breast Cancer with $d=1554$, $n=512$ yields $\lceil d/n \rceil = 4$ BFV ciphertexts per client. With $N=5$ honest clients, uplink is dominated by 5×4 ciphertext pairs (≈ 1.7 MB/round before rejection). After the Byzantine client is rejected, payload drops to ≈ 1.4 MB. End-to-end wall time $\approx 4-5$ s/round on a laptop CPU (NumPy), acceptable for nightly hospital jobs. Accuracy trajectory $88.6\% \rightarrow 93.0\% \rightarrow 93.9\%$ shows learning continues under threshold decryption and Unruh verification. Full-res PneumoniaMNIST (784 pixels, 25 HE chunks) reaches $\approx 67.5\%$ by round 3 while rejecting the scripted oversized client.

b) Takeaway for eHPWAS/WiMob.: The wireless/mobile systems angle is not “faster ML”, but *budgetable* PQ verifiable aggregation: megabytes per round, seconds of CPU, and explicit trust anchors (KEM, ZKP, threshold shares) that match multi-hospital governance. That is the systems story reviewers asked to see beyond a single synthetic plot.

VIII. CONCLUSION

We revised ZKFL-PQ end-to-end against eHPWAS reviews A–D: formal binding and Unruh-oriented NIZK, threshold BFV with partial decryption, full-vector HE, Classic-128 presets, expanded synthetic baselines including sign-flip, and real medical demonstrations (UCI; MedMNIST path). Using the full six-page IEEE budget, we mapped each review theme to concrete changes (Table I), reported multi-seed metrics and WiMob-relevant payloads, and positioned the system against Beskar without oversell. Artifact: <https://github.com/edlansiaux/pq-zkfl-medical>.

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